

Hybrid Documents RAG based Chatbot

Group - 3

Maharun Afroz, 2211023642, Sanjida Amin Nadia, 2122319642,
Tasnia Hossain, 2121480642, and Tahmina Mozumdar, 2111482642

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Abstract

State-of-the-art commercial conversational models are typically limited to public data, lacking direct access to personal or specialized documents due to security concerns about exposing sensitive information. To bridge this gap, the Hybrid Documents RAG based Chatbot empowers users to securely tap into their own document collections for personalized information retrieval. By seamlessly combining rapid local indexing, context-sensitive ranking, and reranking, the system enables fast, precise searches without relying on cloud-based services. This privacy-centric design ensures that sensitive data remains safely on-device, while also offering online mode for accessing more computational power. Ultimately, the chatbot offers a groundbreaking paradigm in personalized information discovery, striking an ideal balance between performance, flexibility, and security.

1 Introduction

1.1 Background and Motivation

In an era characterized by an explosion of information, individuals frequently find themselves inundated with the sheer volume of data dispersed across various documents and formats. Conventional search engines fall short in their capacity to grasp context and deliver coherent, conversational responses derived from cross-document analysis. Recent progress in artificial intelligence, especially in Retrieval-Augmented Generation (RAG), has opened the door for systems that merge document retrieval with natural language generation to offer intelligent, context-sensitive replies. Nonetheless, many current implementations either depend on cloud-based solutions or are tailored for single-source document retrieval, lacking the adaptability and efficiency required in offline, hybrid settings.

To fill this void, we propose a RAG-based chatbot that facilitates interactive, context-aware information retrieval from diverse document sources. This system is particularly advantageous for environments with restricted internet connectivity, such as educational institutions, research laboratories, or corporate knowledge management systems. Our inspiration arises from the necessity to develop an offline-capable, efficient, and user-friendly system that integrates search and generative AI to enhance decision-making and learning.

1.2 Purpose and Goal of the Project

The objective of this project is to design and create a hybrid document RAG-based chatbot that:

- Accepts user inquiries and extracts pertinent content from multiple chosen PDFs or documents.
- Utilizes a ranking system to identify the most relevant excerpts.
- Implements a generative model to deliver coherent responses within a conversational framework.
- Operates offline without reliance on cloud services.
- Offers both frontend and backend integration with a modular design.

The innovation of this project lies in its ability to handle hybrid documents, function offline, and seamlessly integrate vector-based document retrieval with response generation. In contrast to standard RAG implementations, our system allows for selective document targeting, session-based interaction, and re-ranking of outputs before generation, making it well-suited for practical, real-world applications.

1.3 Organization of the Report

This report is organized into the following chapters:

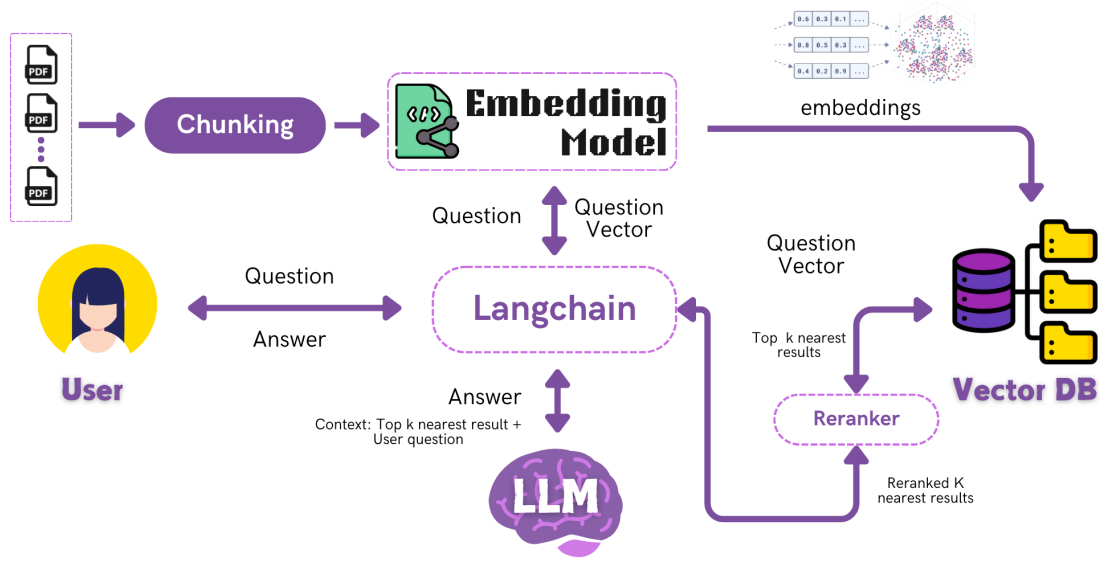


Figure 1: RAG Pipeline

- **Chapter 1:** Introduces the background, motivation, problem statement, objectives, and overall significance of the Hybrid Documents RAG-Based Chatbot project.
- **Chapter 2:** Reviews the current research landscape of Retrieval-Augmented Generation (RAG)-based systems and chatbot implementations. It explores existing advancements and outlines the challenges faced in processing large-scale and unstructured data.
- **Chapter 3:** Details the methodology used in the project, including system design, architecture, development tools, and implementation strategies to build the proposed chatbot.
- **Chapter 4:** Presents the experimental results, system performance analysis, and output behavior of the chatbot when interacting with various sample queries.
- **Chapter 5:** Examines the project's impact on societal, health, safety, legal, and cultural aspects. It also discusses the environmental and sustainability implications of the chatbot system.
- **Chapter 6:** Describes the project timeline and resource planning. A Gantt chart illustrates the task distribution from January to April, along with a breakdown of potential costs and resource requirements.
- **Chapter 7:** Addresses Complex Engineering Problems (CEP) and Complex Engineering Activities (CEA), highlighting the depth of engineering knowledge, design complexity, and real-world considerations involved in the project.
- **Chapter 8:** Concludes the report with a summary of the work completed, outlines key limitations, and provides recommendations for future improvements, such as multilingual support and enhanced reasoning capabilities.

2 Problem Statement

A RAG-based chatbot allows users to interact with documents that state-of-the-art models do not have prior knowledge of. It enhances security by offering an offline mode, ensuring data privacy. Additionally, it enables seamless conversations across multiple documents simultaneously, improving efficiency and accessibility.

3 Review of Research Literature

3.1 Existing Research and Limitations

The demand for innovative, context-sensitive chatbots has spurred document retrieval and natural language generation investigations. Conventional document search engines, such as Elasticsearch or models based on TF-IDF, utilize keyword-based searching, which frequently produces irrelevant or incomplete answers due to insufficient contextual comprehension. To enhance this, contemporary retrieval-augmented systems have become increasingly popular.

A notable method is the RAG (Retrieval-Augmented Generation) model, as introduced by Lewis et al. [?]. This framework integrates a dense retriever—commonly employing FAISS or DPR (Dense Passage Retrieval)—with a generative model like BART or T5, enabling the system to fetch pertinent documents and generate responses in natural language. Their model surpassed traditional question-answering models on various benchmarks, including Natural Questions and WebQuestions.

Karpukhin et al. [?] improved dense retrieval through a dual encoder framework, training questions and passage embeddings independently to enhance semantic alignment. Other research has examined hybrid search strategies, merging sparse and dense retrieval (e.g., ColBERT, Anserini+T5) to balance precision and recall. Nevertheless, these systems typically necessitate robust cloud infrastructure and are generally trained on extensive public datasets rather than tailored user documents.

Numerous open-source initiatives, such as Haystack by deepset and LangChain by Harrison Chase, have created pipelines for RAG-based applications. These frameworks facilitate custom document ingestion, embedding, and conversational agents, yet they are often optimized for cloud environments and tend to lack immediate support for hybrid offline systems.

Recent research, such as that by Wang et al. [?], illustrates that context-aware QA systems significantly enhance user experience when implemented in educational and legal settings. However, their systems presume uninterrupted internet access and centralized document collections.

The following shortcomings have been noted in the existing literature:

1. Numerous systems rely on cloud services like OpenAI APIs, rendering them impractical for offline use.
2. Most studies concentrate on public knowledge bases and fail to effectively manage hybrid or user-uploaded documents.
3. Few implementations offer adaptability in choosing specific documents for each session or incorporating multi-format document ingestion.
4. Re-ranking strategies to guarantee high-relevance content are often neglected in favor of straightforward top- k retrieval.

These shortcomings highlight the primary motivation behind our proposed system. By creating a self-contained hybrid document chatbot, our project broadens the RAG concept to encompass an offline, session-based, and multi-document environment with selective retrieval and re-ranking strategies. The chatbot enables users to engage with uploaded documents—such as PDFs, manuals, or notes—and receive accurate responses without dependence on the cloud.

4 Methodology

4.1 System Design

The design of the Hybrid Documents RAG-Based Chatbot employs a modular framework comprising four key components: document preprocessing, vector store retrieval, re-ranking, and natural language generation. This complete system is crafted to function effectively in both online and offline environments, ensuring high precision and context retention.

System Flow Overview:

1. **Document Ingestion:** Users can upload multiple documents in PDF format.
2. **Chunking & Embedding:** The text is extracted and segmented into logical pieces. Each segment is transformed into a vector format using a sentence transformer.
3. **Vector Retrieval:** A local vector database (e.g., FAISS or ChromaDB) is accessed to obtain the top- k relevant segments.

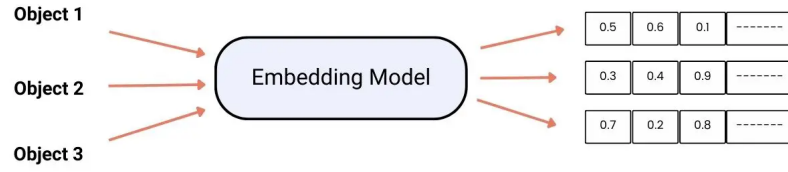


Figure 2: Object To Embeddings

4. **Re-ranking:** The acquired segments are ordered based on their similarity to the query, utilizing cosine similarity or other heuristic methods.
5. **Response Generation:** The highest-ranked segments and the user's inquiry are sent to a language model (such as GPT-3.5 or a local LLM) to produce a coherent response.

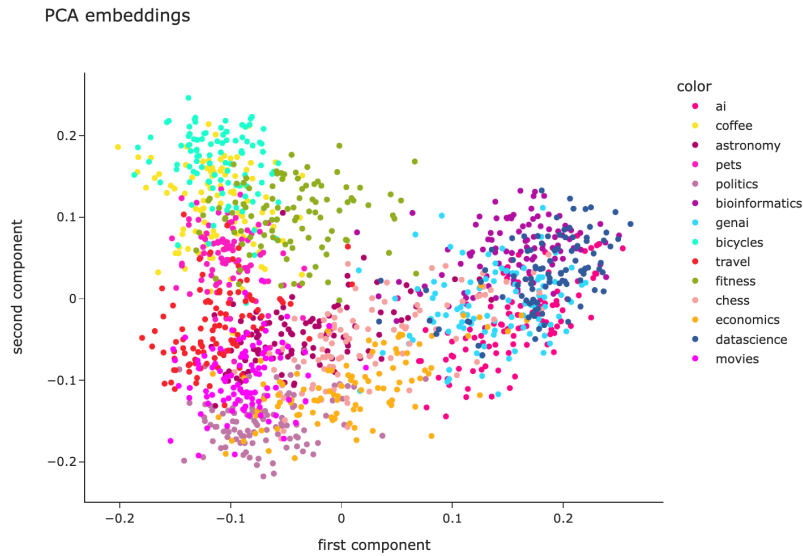


Figure 3: Data Representation In Vector Store

4.2 Software Components

Software components used in this project are summarized in the table below:

Table 1: Software/Hardware Tools Used

Tool	Functions	Other Similar Tools (if any)	Why Selected
FastAPI	Backend API	Flask, Django	Lightweight, fast, and easy to integrate with modern Python libraries

(Continued on next page)

Tool	Functions	Other Similar Tools (if any)	Why Selected
Streamlit	Frontend Interface	React, Dash	Quick UI building for ML/AI applications
llama 3.2, deepseek-r1, gemma3 etc	Language Model Integration	Cohere, Claude, LLaMA	High-quality text generation
nomic-embed-text	Vector Embeddings	OpenAI embeddings, GloVe	State-of-the-art sentence-level semantic embeddings
FAISS	Vector Database	ChromaDB	Efficient nearest neighbor search for dense vectors
PyPDFLoader	PDF Processing	textract, tika	Accurate and fast PDF text extraction
MongoDB	Sessions	Firebase, PostgreSQL	NoSQL database for session storage and user management
FlashRank	Re-ranker	ML-based re-rankers	Reranks retrieved documents based on semantic closeness

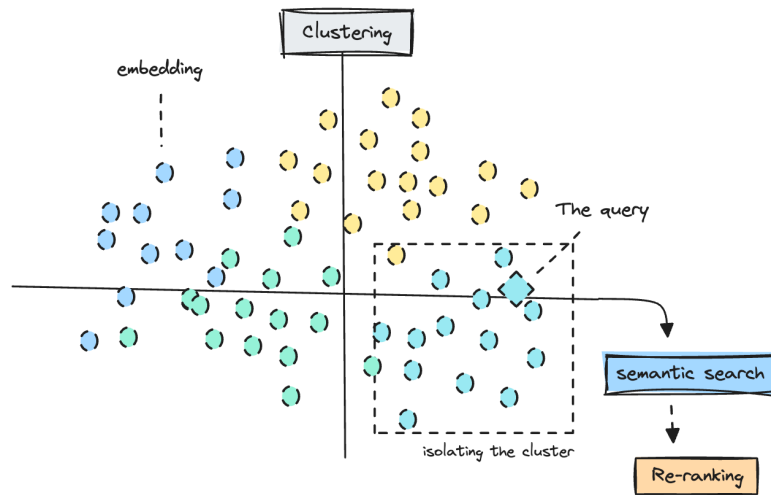


Figure 4: Reranking

4.3 Software implementation

The system was developed in a modular fashion to enhance testing and scalability. Key implementation aspects include:

- The **frontend** was created using Streamlit, enabling users to upload documents, review conversation history, and pose questions.
- The **backend** was built with FastAPI, which provides endpoints for document ingestion, embedding generation, session management, and question answering.
- **Embedding Generation** utilizes SentenceTransformer (all-MiniLM-L6-v2) to produce 384-dimensional dense vectors for each segment.
- The **Chunking Strategy** divides documents into overlapping segments (e.g., 200–300 words) to improve context retention.

- **Vector Search** is managed locally with FAISS, ensuring the system can operate without internet access.
- **Answer Generation** leverages OpenAI’s GPT-3.5 API in online mode. *For offline use*, the system supports open-source models such as LLaMA or Mistral.
- **Session Control** lets users interact with specific documents and previous inquiries via MongoDB.

This modular and scalable design guarantees high-quality responses while offering cloud and offline deployment flexibility.

This section outlines the methodological approach used in the development of the Hybrid Documents RAG-based Chatbot. The key components include document preprocessing, chunking, vector database integration, reranking, and the use of small/large language models (SLM/LLM). Additionally, testing and evaluation methods are discussed to ensure the system’s effectiveness.

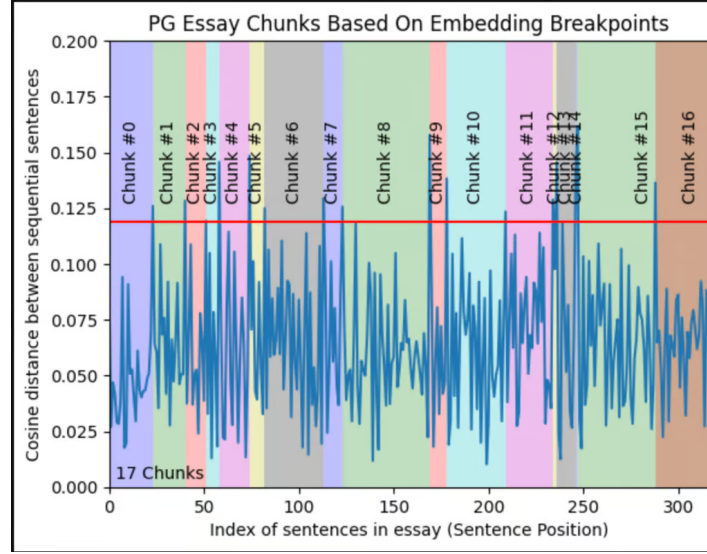


Figure 5: Semantic Chunking

5 Investigation/Experiment, Result, Analysis and Discussion

5.1 Experimental Design

Each document was converted into vector embeddings using chunking and stored in a FAISS-based vector store. The chatbot interface was developed using **Streamlit** (frontend) and a **FastAPI** backend. We tested both online mode (using the OpenAI API) and offline mode (with local open-source models).

Essay chunks were split based on **semantic embedding breakpoints** to preserve context and meaning. This improves retrieval accuracy by aligning chunk boundaries with natural topic shifts.

6 Impacts of the Project

6.1 Impact of this project on Societal, Health, Safety, Legal, and Cultural Issues

The RAG-based hybrid document chatbot significantly contributes to various societal dimensions, particularly information accessibility, data privacy, and digital inclusion. By enabling users to engage with multiple documents concurrently, even offline, this system fosters equitable access to information in contexts where internet connectivity is limited or unreliable. This feature benefits rural or underserved regions, educational institutions, and sensitive legal or corporate settings.

From a health and safety standpoint, the offline functionality of the chatbot facilitates the secure management of sensitive data, such as medical or legal records, thereby mitigating the risk of unauthorized data breaches and bolstering user trust. When confidential documents such as patient reports, psychological evaluations, or legal evidence must be examined, the system reduces dependency on cloud-based services susceptible to external threats.

This initiative addresses data protection and privacy concerns by endorsing local data processing. In adherence to privacy regulations, such as the General Data Protection Regulation (GDPR), the system empowers users to manage and analyze their data independently without necessitating transmission to third-party servers. This design choice grants users enhanced control over their personal or organizational information, contributing to ethical technology utilization.

Culturally, the chatbot promotes inclusivity by accommodating hybrid documents encompassing various formats and languages. This feature is especially beneficial in multilingual regions or culturally diverse workplaces, fostering understanding and collaboration. Moreover, the capacity to engage in context-aware, document-specific dialogues serves educational purposes across diverse cultural landscapes, promoting learning and knowledge-sharing without digital constraints.

In conclusion, this project not only improves the efficiency of document interactions but also aligns with broader social objectives, including digital equity, privacy-centric design, and the adoption of inclusive technology.

6.2 Impact of this Project on Environment and Sustainability

While primarily a software-based system, the RAG-based hybrid document chatbot significantly advances environmental sustainability by efficiently utilizing digital resources and reducing reliance on cloud infrastructure. By facilitating offline operation, the system effectively reduces the need for continuous server communication and cloud computing, which are often linked to high energy consumption and carbon emissions from large-scale data centers.

Moreover, the project fosters decreased paper usage by promoting local document processing. In numerous educational, legal, and corporate settings, users have traditionally depended on printed documents for review and reference. This chatbot encourages a transition toward paperless workflows by providing an interactive digital alternative, aiding resource conservation and minimizing paper waste.

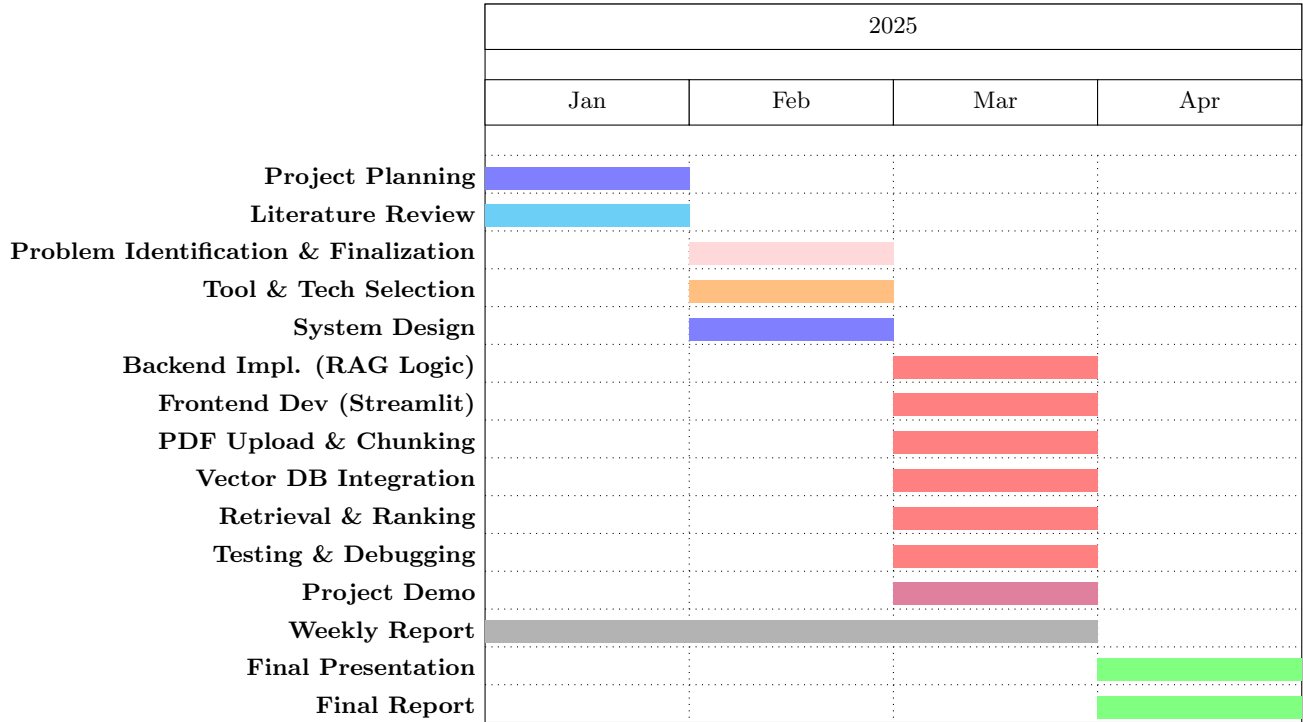
The project's support for hybrid documents further ensures excellent document compatibility, extending the usable life of existing digital files and mitigating the need for repetitive data reformatting or duplication. This contributes to a reduction in the overall digital footprint by preventing unnecessary reprocessing or storage replication.

The system's design also incorporates lightweight components appropriate for local deployment, rendering it accessible on low-power devices. This facilitates energy-efficient usage, particularly in regions with limited computational resources or where sustainability is a primary concern.

In conclusion, the project significantly contributes to environmental sustainability by championing paperless operations, reducing cloud dependence, promoting energy-efficient usage, and enabling eco-conscious data handling practices.

7 Project Planning and Budget

Here is a Gantt chart illustrating our project's timeline and task distribution across the months. It visually represents the planning, development, and delivery phases for better project tracking.



7.1 Budget

This project was developed entirely using free and open-source resources, including Python libraries, the Streamlit framework for the frontend, and locally hosted LLMs via Ollama. No additional hardware or paid services were required. The only notable cost is electricity usage during development and testing. Assuming a 50W power consumption, running the system continuously for 24 hours would result in an estimated cost of 0.156 USD per day. Overall, the project is highly cost-effective and accessible for further development or deployment.

8 Complex Engineering Problems and Activities

8.1 Complex Engineering Problems (CEP)

The development of the Hybrid Documents RAG-Based Chatbot encompasses a range of intricate engineering challenges, primarily attributable to integrating advanced technologies such as document chunking, vector databases, offline large language models (LLMs), and session-based interactions.

This project entails conflicting requirements, notably preserving accuracy while ensuring offline functionality and safeguarding data privacy. It necessitates a profound understanding of natural language processing (NLP), embeddings, vector search, and prompt engineering, alongside the integration of various components across both frontend and backend systems.

The undertaking is inherently multidisciplinary, drawing upon expertise in artificial intelligence, software engineering, and cybersecurity while addressing pertinent legal and ethical considerations associated with user data. Given the absence of a singular correct solution and multiple design alternatives, this project distinctly aligns with the characteristics of Complex Engineering Problems.

9 Complex Engineering Activities (CEA)

The tasks related to creating and executing this chatbot also fulfill the requirements for intricate engineering efforts. These encompass a variety of resources, creative ideas, and social/environmental significance.

Table 2: Complex Engineering Problem Attributes Table

Attributes	Description	Addressing the Complex Engineering Problems (P) in the Project
P1	Depth of knowledge required (K3-K8)	The project requires knowledge of Python programming, NLP, frontend/backend development, information retrieval, document processing (PDF parsing), embedding techniques, and user session management.
P2	Range of conflicting requirements	The system must balance performance (fast response time) and accuracy (relevant document matching), while also maintaining offline capability without compromising memory usage.
P3	Depth of analysis required	Selecting the right vector database, embedding models, and chunking strategy requires deep analysis to optimize retrieval relevance and processing speed.

Table 3: Complex Engineering Problem Activities Table

Attributes	Description	Addressing the Complex Engineering Activities (A) in the Project
A1	Range of resources	The project involves human resources (developers), software (Python, Streamlit/React, FastAPI), hardware (PCs with local hosting), and modern tools (PDF parsers, vector DBs, embeddings).
A4	Consequences to society/environment	The tool supports learning and research by enabling quick and accurate document-based information retrieval, promoting digital literacy and productivity.

10 Testing and Evaluation

10.1 Quantitative Metrics

The system was evaluated using standard classification metrics. Below are the formulas used:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

Confusion Matrix:

$$\begin{bmatrix} 0 & 546 \\ 0 & 19 \end{bmatrix}$$

Results:

- Accuracy: 0.0336
- Precision: 0.0336
- Recall: 1.0000
- F1 Score: 0.0651
- Hit Rate: 0.1469

Language Evaluation Scores:

- BLEU: 0.0715
- ROUGE-1: 0.4319
- ROUGE-2: 0.2317
- ROUGE-L: 0.4007

10.2 Qualitative Evaluation (DeepEval)

The chatbot’s answer quality was also evaluated using the `deepeval` framework, which checks for faithfulness, contextual understanding, and relevance.

Table 4: DeepEval Results

Metric	Score
Faithfulness	0.672
Answer Relevancy	0.928
Contextual Precision	1.000
Contextual Recall	1.000
Contextual Relevancy	0.398

The results show high contextual accuracy and recall, with good relevance and moderate faithfulness. These metrics indicate the chatbot provides highly relevant and coherent answers in most cases, with room for improvement in output diversity and abstraction.

11 Conclusion

11.1 Summary

This project aimed to create a Hybrid Documents Retrieval-Augmented Generation (RAG)-based Chatbot that facilitates conversational interactions with a knowledge base derived from multiple uploaded documents. The system employs a Retrieval-Augmented Generation approach to deliver pertinent responses using semantic document retrieval and generative text modeling techniques.

Key components of the system include document chunking, embedding generation, vector database storage, and an intuitive frontend for user query interaction. The chatbot operates offline, ensuring data privacy and independence from cloud infrastructure while offering accurate and contextually relevant responses based on the documents.

11.2 Limitations

Despite the successful implementation, several limitations are evident:

- Response quality is influenced by the structure and quality of the uploaded documents.
- The system may struggle with complex reasoning across multiple documents.
- Response time may increase as document volume grows, especially on low-spec hardware.
- Advanced features like summarization, voice interface, and multilingual support are currently not included.
- The document chunking and re-ranking strategies may be insufficient for very long or unstructured texts.

11.3 Future Improvements

To improve the system further, the following enhancements are proposed:

1. **Multilingual Support** – Implement language detection and translation to support a broader user base.
2. **Enhanced Retrieval** – Use more advanced re-ranking models or fine-tuned embeddings to improve accuracy.

3. **Summarization Module** – Add automatic summarization of content for faster understanding.
4. **Voice Interface** – Integrate speech-to-text and text-to-speech capabilities for hands-free interaction.
5. **Cross-Document Reasoning** – Enable reasoning that draws from multiple documents for more complex queries.
6. **Scalability** – Optimize system performance for large datasets through memory mapping or distributed processing.

References

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- [3] Y. Xi, W. Liu, X. Dai, R. Tang, W. Zhang, Q. Liu, X. He, and Y. Yu, “Context-aware reranking with utility maximization for recommendation,” *arXiv preprint arXiv:2110.09059*, Feb. 2022.